

SUSHANT BODADE

☎ +91 8530178038 | ✉ sushant.bodade22@spit.ac.in | 🌐 sushant-bodade | 🐙 sushant-bodade | 🏠 Sushant_Bodade

TECHNICAL SKILLS

Languages: Java, SQL, JavaScript, Python, HTML/CSS

Frameworks & Libraries: Selenium, PyTorch, LangChain, FastAPI, React.js, Tailwind CSS, Streamlit

Databases & Vector Stores: MySQL, MongoDB

Tools & Platforms: GitHub, Tableau, VS Code

PROJECTS

BrainSegNet – Brain Tumor Segmentation | PyTorch, Streamlit, FastAPI, Hugging Face, NumPy, Pandas 🌀 **June 2025**

- Achieved a **93% tumor detection accuracy**, **0.87 Dice score**, and **0.17 test loss** on a test set of **531 samples** from the LGG scan dataset by designing and training a custom **U-Net** in **PyTorch** for pixel-wise segmentation of brain tumor regions in MRI scans.
- **Improved generalization** and reduced **overfitting** through extensive data augmentation with **Albumentations**, addressing severe class imbalance.
- Processed **3,900+ MRI slices** across training, validation, and test sets to ensure robust generalization and reproducibility of segmentation results.
- Developed a **FastAPI** backend for MRI tumor segmentation and integrated a **Streamlit UI** for local, interactive visualization.
- Uploaded the trained model to **Hugging Face Hub** (🌀) for public access and reproducibility.

ClipQ – YouTube RAG Assistant | LangChain, Gemini, HuggingFace, FastAPI, Streamlit, Chroma 🌀 **June 2025**

- **Enabled paste-a-link Q&A** over any YouTube video: fetch captions via `youtube-transcript-api`, chunk and embed them, and store vectors in **Chroma**.
- **Delivers answers** by retrieving the **top-k relevant chunks** and prompting **Gemini** to generate responses.
- Exposed the pipeline through clean **FastAPI** endpoints and a minimalist **Streamlit UI** so users can paste a URL, ask questions, and view answers instantly.

VortexAI: A virtual Interview preparation platform | React, NLP, FastAPI, Python 🌀 **Feb 2025 – Present**

- Conceptualized and currently developing VortexAI, a virtual interview preparation application that simulates real-world interview experiences with AI-driven scenarios.
- Engineered a dynamic virtual interviewer using Natural Language Processing (NLP) models to analyze candidate responses and provide real-time feedback.
- Implementing core functionalities through seamless API integration with a FastAPI backend to deliver AI-driven performance analytics.
- Focusing on a robust UI/UX design to provide intuitive navigation and a streamlined user journey through the platform.

RELEVANT COURSEWORK

- | | | | |
|--------------------------------|--------------------|---------------------|-------------------------------|
| • Data Structures & Algorithms | • OOPS | • Operating Systems | • Computer Networks |
| • Database Management | • Machine Learning | • Data Analytics | • Natural Language Processing |

EDUCATION

Sardar Patel Institute of Technology

Bachelor of Technology in Computer Engineering – CGPA : 8.07(till 6th Semester)

Nov 2022 – June 2026

Andheri West, Mumbai

S M College Muktainagar, Jalgaon

HSC - 12th Science – Grade : 89.17%

June 2020 – May 2022

Muktainagar, Jalgaon

S B Chaudhari High School

SSC - 10th Standard – Grade : 86.60%

June 2019 – June 2020

Muktainagar, Jalgaon